

Combination Numbers
- Pascal’s Rule: $C_n^k = C_{n-1}^k + C_{n-1}^{k-1}$. - Absorption Identity: $kC_n^k = nC_{n-1}^{k-1}$. - Trinomial Revision: $C_n^m C_m^k = C_n^k C_{n-k}^{m-k}$. - Row Sum: $\sum_{k=0}^n C_n^k = 2^n$. - Even-Odd Sums: $\sum_{k \geq 0} C_n^{2k} = \sum_{k \geq 0} C_n^{2k+1} = 2^{n-1}$. - Weighted Sums: $\sum_{k=0}^n k C_n^k = n2^{n-1}$. $\sum_{k=0}^n k^2 C_n^k = n(n+1)2^{n-2}$. $\sum_{k=0}^n k^3 C_n^k = n^2(n+3)2^{n-3}$. $\sum_{k=0}^m (-1)^k C_n^k = (-1)^m C_{n-1}^{m-1}$. - Hockey-Stick Identity: $\sum_{i=k}^n C_i^k = C_{n+1}^{k+1}$. - Diagonal Sum: $\sum_{k=0}^m C_{n+k}^k = C_{n+m+1}^{n+m+1}$. - Vandermonde’s Identity: $\sum_{k=0}^r C_n^k C_{n-m}^{r-k} = C_r^{n+m}$.
Probability Space
Sample Space Ω: the set of all possible outcomes. σ-algebra: the collection of events. $\emptyset \in \mathcal{F}$, - If $A \in \mathcal{F}$, then $A^0 \in \mathcal{F}$, - If $A_i \in \mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$. Borel field: all intervals and σ-algebra generated from them. Probability Measure $\mathbf{P}$: a function $\mathbf{P}: \mathcal{F} \rightarrow [0, 1]$. - Nonnegativity: $\mathbf{P}(A) \geq 0, \forall A \in \mathcal{F}$, - Normalization: $\mathbf{P}(\emptyset) = 0, \mathbf{P}(\Omega) = 1$, - Countable additivity: If $A_i \in \mathcal{F}$ are <i>disjoint</i>, $\mathbf{P}\left(\bigcup_i A_i\right) = \sum_i \mathbf{P}(A_i)$.
Conditioning and Independence
Conditioning Definition: For $\mathbf{P}(B) \neq 0, \mathbf{P}(A B) = \mathbf{P}(A \cap B)/\mathbf{P}(B)$. - Nonnegativity: $\mathbf{P}(A B) \geq 0, \forall A \in \mathcal{F}$, - Normalization: $\mathbf{P}(\Omega B) = 1$, - Countable additivity: If $A_i \in \mathcal{F}$ are <i>disjoint</i>, $\mathbf{P}\left(\bigcup_i A_i B\right) = \sum_i \mathbf{P}(A_i B)$. Multiplication Rule: $\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B A) \cdot \mathbf{P}(C A \cap B)$. Total Probability Theorem: For a partition of sample space $A_1, A_2, \dots, A_n$, assuming $\mathbf{P}(A_i) > 0$ for all i, $\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(A_i)\mathbf{P}(B A_i).$ Bayes’s Rule: $\mathbf{P}(A_i B) = \frac{\mathbf{P}(A_i)\mathbf{P}(B A_i)}{\sum_{j=1}^n \mathbf{P}(A_j)\mathbf{P}(B A_j)}$. Independence Definition: Event A and B are called independent if $\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$. - For $\mathbf{P}(B) > 0, \mathbf{P}(A B) = \mathbf{P}(A)$. Conditional Independence: $\mathbf{P}(A \cap B C) = \mathbf{P}(A C)\mathbf{P}(B C)$. Collective Independence: Events $A_1, A_2, \dots, A_n$ are called independent if $\mathbf{P}(A_i \cap A_j \cap \dots \cap A_q) = \mathbf{P}(A_i)\mathbf{P}(A_j) \dots \mathbf{P}(A_q),$ for any distinct indices $i, j, \dots, q$ chosen from $\{1, 2, \dots, n\}$. Inclusion-Exclusion Principle: $\mathbf{P}\left(\bigcup_{i=1}^n A_i\right) = \sum \mathbf{P}(A_i) - \sum_{i < j} \mathbf{P}(A_i \cap A_j) + \dots + (-1)^{n-1} \mathbf{P}(A_1 \cap \dots \cap A_n)$. Hat Problem (Derangement): - Derangement Number: $D_n = n! \sum_{i=0}^n (-1)^i / i!$ $D_n = nD_{n-1} + (-1)^n.$ $\lim_{n \rightarrow \infty} D_n/n! = 1/e.$ - Prob. of at least one match is $\approx 1 - 1/e$ as $n \rightarrow \infty$. Symmetric Difference: $\mathbf{P}(A \triangle B) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)$.

Discrete Random Variable
Random Variable: Given a probability space $(\Omega, \mathcal{F}, \mathbf{P})$, a random variable is a <i>function</i> $X: \Omega \rightarrow \mathbb{R}$ such that $X^{-1}(A) \in \mathcal{F}$ for each $A \in \mathcal{B}(\mathbb{R})$. PMF: $p_X(x) = \mathbf{P}(X = x)$. Derived PMF: $Y = g(X) \Rightarrow p_Y(y) = \sum_{x: g(x)=y} p_X(x)$. Expectation: $\mathbf{E}[X] = \sum_i x_i p_X(x_i)$. Variance: $\text{var}[X] = \mathbf{E}\left[(X - \mathbf{E}[X])^2\right] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$. $\mathbf{E}[\alpha X + \beta] = \alpha \mathbf{E}[X] + \beta, \mathbf{E}[X + Y] = \mathbf{E}[X] + \mathbf{E}[Y]$. $\text{var}[\alpha X + \beta] = \alpha^2 \text{var}[X]$. Geometric RV: $p_X(k) = (1 - p)^{k-1} p$, $\mathbf{E}[X] = 1/p$, $\mathbf{E}[X^2] = (2 - p)/p^2$, $\text{var}[X] = (1 - p)/p^2$. Memorylessness: $p_X(k X > n) = (1 - p)^{k-n-1} p; \mathbf{E}[X X > n] = 1/p + n$. Bernoulli RV: $p_X(k) = \begin{cases} p, & k = 1, \\ 1 - p, & k = 0, \end{cases}$ $\mathbf{E}[X] = p$, $\mathbf{E}[X^2] = p$, $\text{var}[X] = p(1 - p)$. Discrete Uniform RV: $p_X(k) = 1/(b - a + 1)$, if $k = a, a + 1, \dots, b$, $\mathbf{E}[X] = (a + b)/2$, $\text{var}[X] = (b - a)(b - a + 2)/12$. Binomial RV: $p_X(k) = C_n^k p^k (1 - p)^{n-k}$, $\mathbf{E}[X] = np$, $\mathbf{E}[X^2] = np(1 - p + np)$, $\text{var}[X] = np(1 - p)$. Poisson RV: $\lambda = np$ as $n \rightarrow \infty$ in binomial RV. $p_X(k) = e^{-\lambda} \lambda^k / k!, k = 0, 1, \dots$, $\mathbf{E}[X] = \lambda$, $\mathbf{E}[X^2] = \lambda + \lambda^2$, $\text{var}[X] = \lambda$, - Independent sum: $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.
Multiple Discrete RV
Conditioning: Conditional PMF: $p_{X A}(x) = \mathbf{P}(\{X = x\} \cap A)/\mathbf{P}(A), p_{X Y}(x y) = p_{X,Y}(x, y)/p_Y(y)$. Conditional Expectation: $\mathbf{E}[X A] = \sum_x x p_{X A}(x)$. Total Expectation Theorem: $\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X A_1] + \dots + \mathbf{P}(A_n)\mathbf{E}[X A_n]$. Marginal PMF: $p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z); p_X(x) = \sum_y p_{X,Y}(x, y)$. Independence: $\forall x, y: p_{X,Y}(x, y) = p_X(x)p_Y(y)$. - For Independent RVs: $f(X)$ and $g(Y)$ are also independent. $\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$. $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y), \text{var}(X - Y) = \text{var}(X) + \text{var}(Y)$. Convolution: $W = X + Y \Rightarrow p_W(w) = \sum_x p_X(x)p_Y(w - x)$. Conditional Independence: $\forall x, y: p_{X,Y}(x, y A) = p_X(x A)p_Y(y A)$.
Continuous RV
PDF: $\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) \, dx;$ $\mathbf{P}(x \leq X \leq x + \delta) \approx f_X(x)\delta$. Expectation: $\mathbf{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) \, dx$. Variance:

$\text{var}(X) = \int_{-\infty}^{+\infty} (x - \mathbf{E}[X])^2 f_X(x) \, dx = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$.
Uniform RV: $f_X(x) = 1/(b - a), a \leq x \leq b$, $\mathbf{E}[X] = (a + b)/2$, $\mathbf{E}[X^2] = (a^2 + ab + b^2)/3$, $\text{var}[X] = (b - a)^2/12$. Exponential RV: $f_X(x) = \lambda e^{-\lambda x}, x \geq 0$, $F_X(x) = 1 - e^{-\lambda x}, x \geq 0$. $\mathbf{E}[X] = 1/\lambda$, $\mathbf{E}[X^2] = 2/\lambda^2$, $\text{var}[X] = 1/\lambda^2$, $\mathbf{P}(X_1 < X_2) = \lambda_1/(\lambda_1 + \lambda_2)$. - Memorylessness: $f_X(X > a) = \lambda e^{-\lambda(x-a)}, \mathbf{P}(X \geq c + x X \geq c) = \mathbf{P}(X \geq x) = e^{-\lambda x};$ $\mathbf{E}[X X > c] = 1/\lambda + c$. Gaussian RV ($X \sim N(\mu, \sigma^2)$): $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)},$ $F_X(x) = \Phi(x), \Lambda \sim N(0, 1).$ $\mathbf{E}[X] = \mu$, $\mathbf{E}[X^2] = \mu^2 + \sigma^2$, $\mathbf{E}[X^4] = 3\sigma^4 + 6\sigma^2\mu^2 + \mu^4$, $\text{var}[X] = \sigma^2$. $Y = \alpha X + \beta \sim N(\alpha\mu + \beta, \alpha^2\sigma^2)$. CDF: $F_X(x) = \int_{-\infty}^x f_X(t) \, dt$. $F_X(x)$ discontinuous $\Rightarrow X$ has discrete part. $\mathbf{P}(a < X \leq b) = F_X(b) - F_X(a)$. $\mathbf{E}[X] = \int_0^{+\infty} (1 - F_X(x)) \, dx - \int_{-\infty}^0 F_X(x) \, dx$. Derived PDF: CDF Method: find $F_Y(y) = \mathbf{P}(g(X) \leq y) \Rightarrow f_Y(y) = F_Y'(y)$. Linear: $Y = aX + b \Rightarrow f_Y(y) = \frac{1}{ a } f_X\left(\frac{y-b}{a}\right)$. Monotone: $y = g(x) \Leftrightarrow x = h(y) \Rightarrow f_Y(y) = f_X(h(y)) h'(y)$. Min and Max of i.i.d. RVs ($X_1, \dots, X_n$): Max: $Z = \max(X_i) \Rightarrow F_Z(z) = [F_X(z)]^n \implies f_Z(z) = n[F_X(z)]^{n-1} f_X(z)$. Min: $W = \min(X_i) \implies F_W(w) = 1 - [1 - F_X(w)]^n$. <i>Shortcut</i>: If $X_i \sim \text{Exp}(\lambda_i)$, then $\min(X_1, \dots, X_n) \sim \text{Exp}(\sum \lambda_i)$. 2D Uniform & Geometric Probability: $f_{X,Y}(x, y) = 1/\text{Area}(S)$ for $(x, y) \in S$. Broken Stick / Triangle Inequality: Break a line into 3 pieces. They form a triangle iff. all pieces are shorter than 1/2. Prob. is 1/4.

Multiple Continuous RV
Conditioning: Conditional PDF: $f_{X Y}(x y) = f_{X,Y}(x, y)/f_Y(y)$. Conditional Expectation: $\mathbf{E}[X Y = y] = \int_{-\infty}^{+\infty} x f_{X Y}(x y) \, dx$. Total Expectation Theorem: $\mathbf{E}[X] = \int_{-\infty}^{+\infty} \mathbf{E}[X Y = y] f_Y(y) \, dy$. Marginal PDF: $f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) \, dy$. Independence: $\forall x, y: f_{X,Y}(x, y) = f_X(x)f_Y(y)$. - For independent RVs: $f(X)$ and $g(Y)$ are also independent, $\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$. Convolution: If X and Y are independent and $Z = X + Y, f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z - x) \, dx$. $X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2) \Rightarrow X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$. Joint / Multiple CDF: $F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y)$. Marginals: $F_X(x) = F_{X,Y}(x, \infty), F_Y(y) = F_{X,Y}(\infty, y)$. From/To PDF: $F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, dv \, du$ $\implies f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y).$ Independence: $X \perp Y \iff F_{X,Y}(x, y) = F_X(x)F_Y(y)$. Conditional CDF: Given Event A: $F_{X A}(x) = \mathbf{P}(X \leq x \mid A) = \frac{\mathbf{P}(\{X \leq x\} \cap A)}{\mathbf{P}(A)}$.
Given $Y = y$: $F_{X Y}(x y) = \int_{-\infty}^x f_{X Y}(t y) \, dt$. Derivative: $f_{X Y}(x y) = \frac{\partial}{\partial x} F_{X Y}(x y)$. Total Prob: $F_X(x) = \sum \mathbf{P}(A_i)F_{X A_i}(x)$. Continuous Bayes’ Rule: $f_{X Y}(x y) = \frac{f_{Y X}(y x)f_X(x)}{f_Y(y)} = \frac{f_{Y X}(y x)f_X(x)}{\int_{-\infty}^{\infty} f_{Y X}(y t)f_X(t)dt},$
$\mathbf{P}(A Y = y) = \frac{\mathbf{P}(A)f_{Y A}(y)}{f_Y(y)}.$ (Mixed Bayes)
Covariance and Correlation
Covariance: $\text{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])] = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y]$. Uncorrelated: $\text{cov}(X, Y) = 0$; independent $\Rightarrow \text{cov}(X, Y) = 0$. $\text{cov}(X, X) = \text{var}(X); \text{cov}(X, aY + b) = a \text{cov}(X, Y); \text{cov}(X, Y + Z) = \text{cov}(X, Y) + \text{cov}(X, Z)$. $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y) + 2 \text{cov}(X, Y)$. $\text{var}(\sum_i X_i) = \sum_i \text{var}(X_i) + \sum_{i \neq j} \text{cov}(X_i, X_j)$.
Correlation: $\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}, -1 \leq \rho \leq 1$. $\rho = 1 \iff X - \mathbf{E}[X] = c(Y - \mathbf{E}[Y])$.
Conditional Expectation
Iterated Expectation: $\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X Y]]$. Y discrete: $\mathbf{E}[\mathbf{E}[X Y]] = \sum_y \mathbf{E}[X Y = y]p_Y(y)$. Y continuous: $\mathbf{E}[\mathbf{E}[X Y]] = \int \mathbf{E}[X Y = y]f_Y(y) \, dy$. Estimator: $\hat{X} = \mathbf{E}[X Y], \tilde{X} = X - \hat{X}$. $\mathbf{E}[\tilde{X} Y] = 0 \Rightarrow \mathbf{E}[\tilde{X}] = 0; \text{cov}(\hat{X}, \tilde{X}) = 0$. Conditional Variance: $\text{var}(X Y) = \mathbf{E}[(X - \mathbf{E}[X Y])^2 Y]$. Total Variance: $\text{var}(X) = \text{var}(\mathbf{E}[X Y]) + \mathbf{E}[\text{var}(X Y)]$.

Transforms
Definition: $\mathcal{M}_X(s) = \mathbb{E}[\mathrm{e}^{sX}]$; discrete $\sum_x p_X(x) \mathrm{e}^{sx}$; continuous $\int f_X(x) \mathrm{e}^{sx} \, \mathrm{d}x$. Recover Law: MGF exists near 0 $\Rightarrow$ unique distribution. - Discrete: expand $\mathcal{M}_X(s) = \sum_x p_X(x) \mathrm{e}^{sx}$; coeff. of e^{sx} is $p_X(x)$; or $p_X(x) = \mathcal{Z}^{-1}\{\mathcal{M}_X(-\ln z)\}$. - Continuous: $f_X(x) = \mathcal{L}_b^{-1}\{\mathcal{M}_X(-s)\}$; or $f_X(x) = \mathcal{F}^{-1}\{\mathcal{M}_X(it)\} = \frac{1}{2\pi} \int \mathrm{e}^{-itx} \mathcal{M}_X(it) \, \mathrm{d}t$. $\mathcal{M}_X(0) = 1$; $\mathbb{E}[X^n] = \left. \frac{\mathrm{d}^n \mathcal{M}_X(s)}{\mathrm{d}s^n} \right _{s=0}$. $Y = aX + b \Rightarrow \mathcal{M}_Y(s) = \mathrm{e}^{bs} \mathcal{M}_X(as)$. - Independent $Z = \sum_{i=1}^n X_i \Rightarrow \mathcal{M}_Z(s) = \prod_i \mathcal{M}_{X_i}(s)$. Mixture: $f_Y = \sum_i p_i f_{X_i} \Rightarrow \mathcal{M}_Y(s) = \sum_i p_i \mathcal{M}_{X_i}(s)$. Random Sum: $Y = X_1 + \cdots + X_N$, i.i.d. $X_i \perp N$: $\mathbb{E}[Y] = \mathbb{E}[N]\mathbb{E}[X]$, $\mathrm{var}(Y) = \mathbb{E}[N]\mathrm{var}(X) + (\mathbb{E}[X])^2 \mathrm{var}(N)$, $\mathcal{M}_Y(s) = \mathcal{M}_N(\ln \mathcal{M}_X(s))$. Common $\mathcal{M}_X(s)$: - Bernoulli: $1 - p + pe^s$; Binomial: $(1 - p + pe^s)^n$, - Geometric: $\frac{pe^s}{1 - (1 - p)e^s}$; Poisson: $\exp(\lambda(\mathrm{e}^s - 1))$,
- Discrete uniform $\{a, \dots, b\}$: $\frac{\mathrm{e}^{as}(1 - \mathrm{e}^{(b-a+1)s})}{(b - a + 1)(1 - \mathrm{e}^s)}$, - Uniform $[a, b]$: $\frac{\mathrm{e}^{bs} - \mathrm{e}^{as}}{(b - a)s}$; Exponential: $\frac{\lambda}{\lambda - s}$, - Gaussian $N(\mu, \sigma^2)$: $\exp(\mu s + \sigma^2 s^2/2)$.

Probability Inequalities
Markov: $X \geq 0$, $a > 0 \Rightarrow \mathbf{P}(X \geq a) \leq \mathbb{E}[X]/a$. “=” iff $X \in \{0, a\}$. Chebyshev: $\mathbf{P}(X - \mathbb{E}[X] \geq c) \leq \mathrm{var}(X)/c^2$. “=” iff $X \in \{\mu - c, \mu, \mu + c\}$, $\mathbf{P}(X = \mu \pm c) = \sigma^2/(2c^2)$. Variance Bound: $X \in [a, b] \Rightarrow \mathrm{var}(X) \leq (b - a)^2/4$. “=” iff $\mathbf{P}(X = a) = \mathbf{P}(X = b) = 1/2$. Chernoff: - Right-tail: $\mathbf{P}(X \geq a) \leq \inf_{s \geq 0} \mathrm{e}^{-sa} \mathcal{M}_X(s) = \mathrm{e}^{-\sup_{s \geq 0} (sa - \ln \mathcal{M}_X(s))}$, - Left-tail: $\mathbf{P}(X \leq a) \leq \inf_{s \leq 0} \mathrm{e}^{-sa} \mathcal{M}_X(s) = \mathrm{e}^{-\sup_{s \leq 0} (sa - \ln \mathcal{M}_X(s))}$.

LLN and CLT
WLLN: $M_n = \frac{1}{n} \sum_{i=1}^n X_i$, i.i.d., $\mathbb{E}[X_i] = \mu$, $\mathrm{var}(X_i) = \sigma^2$. $\mathbb{E}[M_n] = \mu$, $\mathrm{var}(M_n) = \sigma^2/n$, $\mathbf{P}(M_n - \mu \geq \epsilon) \leq \sigma^2/(n\epsilon^2) \rightarrow 0$, $M_n \xrightarrow{P} \mu$. Convergence: In Probability: $X_n \xrightarrow{P} X \Leftrightarrow \mathbf{P}(X_n - X \geq \epsilon) \rightarrow 0, \forall \epsilon > 0$, In Distribution: $X_n \xrightarrow{D} X \Leftrightarrow F_{X_n}(x) \rightarrow F_X(x)$ at continuous x, Almost Surely: $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow \mathbf{P}(\lim_n X_n = X) = 1$, * “Almost Surely”: For almost all sample paths ω. r-th Mean: $X_n \xrightarrow{r} X \Leftrightarrow \mathbb{E}[X_n - X ^r] \rightarrow 0$, Relations: a.s. $\Rightarrow \mathbf{P} \Rightarrow D$; r-mean $\Rightarrow \mathbf{P} \Rightarrow D$; $r \geq s \Rightarrow L^r \Rightarrow L^s$. CLT: $S_n = \sum_{i=1}^n X_i$, i.i.d., $\mathbb{E}[X_i] = \mu$, $\mathrm{var}(X_i) = \sigma^2$. $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{D} N(0, 1)$; $\mathbf{P}(Z_n \leq z) \rightarrow \Phi(z)$. Transform: $\mathcal{M}_{Z_n}(s) = \mathrm{e}^{-s\mu\sqrt{n}/\sigma} \left(\mathcal{M}_{X_1} \left(\frac{s}{\sigma\sqrt{n}} \right) \right)^n \rightarrow \mathrm{e}^{s^2/2}$. Normal Approx.: $S_n \approx N(n\mu, n\sigma^2)$; $\mathbf{P}(S_n \leq c) \approx \Phi \left(\frac{c - n\mu}{\sigma\sqrt{n}} \right)$. Binomial Approx.: $X_i \sim \text{Bernoulli}(p)$, $S_n = \sum_i X_i \sim \text{Bin}(n, p)$, $q = 1 - p$. $\frac{S_n - np}{\sqrt{npq}} \approx N(0, 1)$, $\mathbf{P}(k \leq S_n \leq l) \approx \Phi \left(\frac{l + 0.5 - np}{\sqrt{npq}} \right) - \Phi \left(\frac{k - 0.5 - np}{\sqrt{npq}} \right)$. SLLN: $M_n = S_n/n \Rightarrow \mathbf{P}(\lim_{n \rightarrow \infty} M_n = \mu) = 1$, i.e. $M_n \xrightarrow{\text{a.s.}} \mu$.

Event Limits and Borel–Cantelli
Set Limits: $\limsup A_n = \bigcap_{k \geq 1} \bigcup_{n \geq k} A_n = \{A_n \text{ i.o.}\}$; * “infinite often” $= \forall N \geq 0, \exists n \geq N, A_n$ happens. $\liminf A_n = \bigcup_{k \geq 1} \bigcap_{n \geq k} A_n = \{A_n \text{ eventually}\}$. * “eventually” $= \exists N \geq 0, \forall n \geq N, A_n$ happens. $\liminf A_n \subseteq \limsup A_n$; “=” $\Rightarrow \lim A_n$ exists. Monotone Convergence Theorem: $0 \leq X_n \uparrow X \Rightarrow \mathbb{E}[X_n] \uparrow \mathbb{E}[X]$; events $A_n \uparrow A \Rightarrow \mathbf{P}(A_n) \uparrow \mathbf{P}(A)$. - Let X_n = the sum of an RV seq. $\{Y_i\}$, $Y_i \geq 0$, $X = \lim_n X_n$, $\mathbb{E}[X] < \infty \Rightarrow \mathbf{P}(X = \infty) = 0 \Rightarrow X$ converge a.s.. $X = \sum_{i=1}^\infty Y_i$ converges a.s. $\Rightarrow \{Y_i\} \xrightarrow{\text{a.s.}} 0$. Borel–Cantelli I: $\sum_n \mathbf{P}(A_n) < \infty \Rightarrow \mathbf{P}(A_n \text{ i.o.}) = 0$; Borel–Cantelli II: independent A_n, $\sum_n \mathbf{P}(A_n) = \infty \Rightarrow \mathbf{P}(A_n \text{ i.o.}) = 1$. a.s. Criterion: $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow \forall \epsilon > 0, \mathbf{P}(X_n - X > \epsilon \text{ i.o.}) = 0$. - Equiv.: $\lim_{n \rightarrow \infty} \mathbf{P}(\sup_{k \geq n} X_k - X > \epsilon) = 0, \forall \epsilon > 0$, - Enough: $\forall m \geq 1, \sum_n \mathbf{P}(X_n - X > 1/m) < \infty$. Subsequence: $X_n \xrightarrow{P} X \Rightarrow \exists n_k : X_{n_k} \xrightarrow{\text{a.s.}} X$; - proof: choose $\mathbf{P}(X_{n_k} - X > 1/k) < 2^{-k}$. Kolmogorov Ineq.: independent RVs X_i, $\mathbb{E}[X_i] = 0$, $\mathrm{var}(X_i) < \infty$, $S_k = \sum_{i=1}^k X_i \Rightarrow \mathbf{P}(\max_{k \leq n} S_k \geq a) \leq \mathrm{var}(S_n)/a^2 = \sum_{i=1}^n \mathrm{var}(X_i)/a^2$. Cantelli SLLN: i.i.d., $\mathbb{E}[X^4] < \infty \Rightarrow \mathbb{E}[(\sum_{i=1}^n (X_i - \mu))^4] = O(n^2)$; Borel–Cantelli I $\Rightarrow S_n/n \xrightarrow{\text{a.s.}} \mu$. - lim sup A_n is event; lim sup $\mathbf{P}(A_n)$ is number; $\mathbf{P}(\limsup A_n) \geq \limsup \mathbf{P}(A_n)$.
Stochastic Processes
Stochastic Process: A collection of RVs indexed by time. View as a mapping: $Z : T \times \Omega \rightarrow \mathbb{R}; (t, \omega) \mapsto Z_t(\omega)$. View as a sequence: $\{Z_{t_1}, Z_{t_2}, \dots\}$, $t_1, t_2 \in T$ is a sequence of time-relevant RVs defined on prob. space $(\Omega, \mathcal{F}, \mathbf{P})$. - Once the sample path ω is fixed, values of the whole RV walk $\{Z_{t_1}(\omega), Z_{t_2}(\omega), \dots\}$ are fixed. Bernoulli Process: discrete-time discrete-value SP $\{X_i\}_{i \geq 1}$; $X_i \sim$ i.i.d. Bernoulli(p). Intuition: keep tossing a coin and record if i-th round is a head as X_i. Independence: seq. $X_{n+1}, X_{n+2}, \dots$ after n is a Bernoulli SP independent of $X_1, \dots, X_n$. Memoryless: W_n = trials from n to next success $\sim \text{Geo}(p)$, Interarrival: Y_k = time of k-th success, $T_1 = Y_1$, $T_k = Y_k - Y_{k-1}$; $T_i \sim$ i.i.d. $\text{Geo}(p)$; $Y_k = \sum_{i=1}^k T_i$, k-th success: $Y_k \sim \text{Pascal}(k, p)$, * $p_{Y_k}(t) = C_{t-1}^{k-1} p^k (1 - p)^{t-k}$, $t = k, k + 1, \dots$, * $\mathbb{E}[Y_k] = k/p$, $\mathrm{var}(Y_k) = k(1 - p)/p^2$, Splitting: keep w.p. $q \Rightarrow \text{Bernoulli SP}(pq)$, Merging Independent Bernoullis: Bernoulli(p) OR Bernoulli(q)$\Rightarrow$ Bernoulli($p + q - pq$). Poisson Process: continuous-time discrete-value SP $\{N_t\}_{t \geq 0}$, $N_0 = 0$, $N_t \sim \text{Poisson}(\lambda t)$; . Intuition: count the number of arrivals w.p. $\lambda \delta$ for a small interval δ, from 0 to t as N_t. Homogeneity: $\mathbf{P}(k, t) = \mathbf{P}(k \text{ arrivals in interval of duration } t)$ $= \sum_{n=0}^\infty \mathbf{P}(N_{t+\tau} = n + k N_\tau = n) \mathbf{P}(n, T)$ Independent: Numbers of arrivals in disjoint time intervals are independent. Small δ: $P(0, \delta) = 1 - \lambda \delta + o(\delta)$, $P(1, \delta) = \lambda \delta + o(\delta)$, $P(k, \delta) = o(\delta)$, $k > 1$, Bernoulli Relation: Select time infinitesimal δ, $n = t/\delta$, $p = \lambda \delta$; $\{X_i\}$ is a Bernoulli SP(p), $S_n = \sum_{k=1}^n X_i \sim \text{Binomial RV}(n, p) \rightarrow N_t \sim \text{Poisson RV } (\lambda t)$, $n \rightarrow \infty$, First Arrival: $Y_1 = T_1 \sim \text{Exp}(\lambda)$; $F_{Y_1}(t) = 1 - \mathrm{e}^{-\lambda t}$, $f_{Y_1}(t) = \lambda \mathrm{e}^{-\lambda t}$; Interarrival: Y_k = time of kth arrival, $T_1 = Y_1$, $T_k = Y_k - Y_{k-1}$; $T_i \sim$ i.i.d. $\text{Exp}(\lambda)$, $Y_k = \sum_{i=1}^k T_i$, k-th arrival: $Y_k \sim \text{Erlang}(\lambda, k)$, * $f_{Y_k}(y) = \lambda^k y^{k-1} \mathrm{e}^{-\lambda y}/(k - 1)!$, $y \geq 0$; * $\mathbb{E}[Y_k] = k/\lambda$, $\mathrm{var}(Y_k) = k/\lambda^2$, Splitting: keep w.p. $p \Rightarrow \text{Poisson}(\lambda p)$, Merging Independent Poissons: $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) =$

Poisson($\lambda_1 + \lambda_2$);
 next arrival from SP 1 w.p. $\lambda_1 / (\lambda_1 + \lambda_2)$.

- **Random Sum:** $Y = \sum_{i=1}^N X_i$, $Y = 0$ if $N = 0$; $\mathcal{M}_Y(s) = \mathcal{M}_N(\ln \mathcal{M}_X(s))$.
- **Random Sum Table:**

X_i	N	Y
Bernoulli(p)	Binomial(m, q)	Binomial(m, pq)
Bernoulli(p)	Poisson(λ)	Poisson(λp)
Geo(p)	Geo(q)	Geo(pq)
Exponential(λ)	Geo(q)	Exponential(λq)

Entropy
Discrete: $H(X) = \mathbb{E}[-\ln p_X(X)] = -\sum_k p_X(x_k) \ln p_X(x_k)$. Continuous: $h(X) = \mathbb{E}[-\ln f_X(X)] = -\int f_X(x) \ln f_X(x) \, \mathrm{d}x$. Bernoulli: $H(X) = -p \log_2 p - (1 - p) \log_2(1 - p)$; max 1 at $p = 1/2$, min 0 at $p = 0, 1$. Max Entropy: - finite $\{x_1, \dots, x_n\} \Rightarrow p_k = 1/n$, $[a, b] \Rightarrow U(a, b)$, $[0, \infty)$, $\mathbb{E}[X] = 1/\lambda \Rightarrow \text{Exp}(\lambda)$, $\mathbb{R}$, fixed $\mu, \sigma^2 \Rightarrow N(\mu, \sigma^2)$.

Integration Formula
$\int x \mathrm{e}^{ax} \, \mathrm{d}x = \frac{1}{a^2} (ax - 1) \mathrm{e}^{ax}$, $\int x^2 \mathrm{e}^{ax} \, \mathrm{d}x = \frac{a^2 x^2 - 2ax + 2}{a^3} \mathrm{e}^{ax}$, $\int x^3 \mathrm{e}^{ax} \, \mathrm{d}x = \frac{a^3 x^3 - 3a^2 x^2 + 6ax - 6}{a^4} \mathrm{e}^{ax}$, $\int x^n \mathrm{e}^{ax} \, \mathrm{d}x = \frac{1}{a} x^n \mathrm{e}^{ax} - \frac{n}{a} \int x^{n-1} \mathrm{e}^{ax} \, \mathrm{d}x$. $\int_0^\infty x^n \mathrm{e}^{-ax} \, \mathrm{d}x = n!/a^{n+1}$. $\int_a^\infty x^n \mathrm{e}^{-\lambda x} \, \mathrm{d}x = \mathrm{e}^{-\lambda a} \sum_{j=0}^n \frac{n! a^j}{j! \lambda^{n+1-j}}$. $\int_a^\infty \lambda \mathrm{e}^{-\lambda x} \, \mathrm{d}x = \mathrm{e}^{-\lambda a}$; $\int_0^a \lambda \mathrm{e}^{-\lambda x} \, \mathrm{d}x = 1 - \mathrm{e}^{-\lambda a}$. $\int_a^\infty \frac{\lambda^k x^{k-1} \mathrm{e}^{-\lambda x}}{(k-1)!} \, \mathrm{d}x = \mathrm{e}^{-\lambda a} \sum_{j=0}^{k-1} \frac{(\lambda a)^j}{j!}$. $\int \sqrt{a^2 - x^2} \, \mathrm{d}x = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{a}$. $\int \ln x \, \mathrm{d}x = x \ln x - x$, $\int x^n \ln x \, \mathrm{d}x = \frac{1}{n+1} x^{n+1} \left(\ln x - \frac{1}{n+1} \right)$. $\int x \exp \left(-x^2/\sigma^2 \right) \, \mathrm{d}x = -\frac{\sigma^2}{2} \exp \left(-x^2/\sigma^2 \right) + C$. $\int_{-\infty}^{+\infty} \exp \left(- (x - \lambda)^2/\mu^2 \right) \, \mathrm{d}x = \mu \sqrt{\pi}$. $\int_{-\infty}^{+\infty} x^2 \exp \left(-x^2/\mu^2 \right) \, \mathrm{d}x = \frac{\mu^3}{2} \sqrt{\pi}$.